

Technology Essentials

Who Am I ?

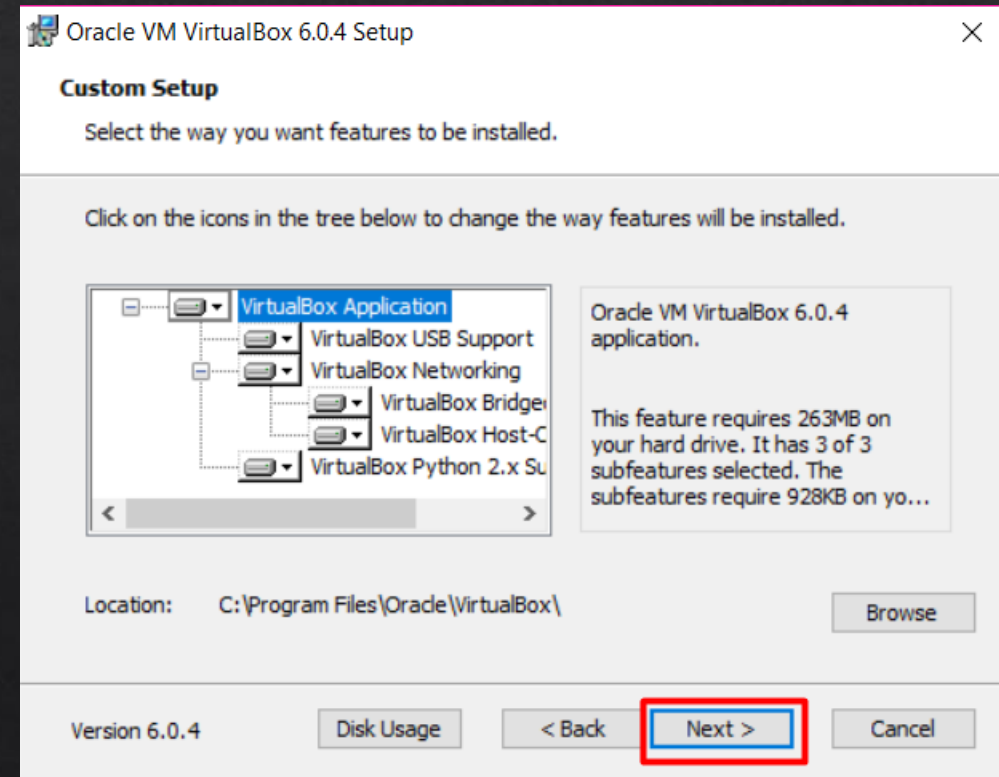
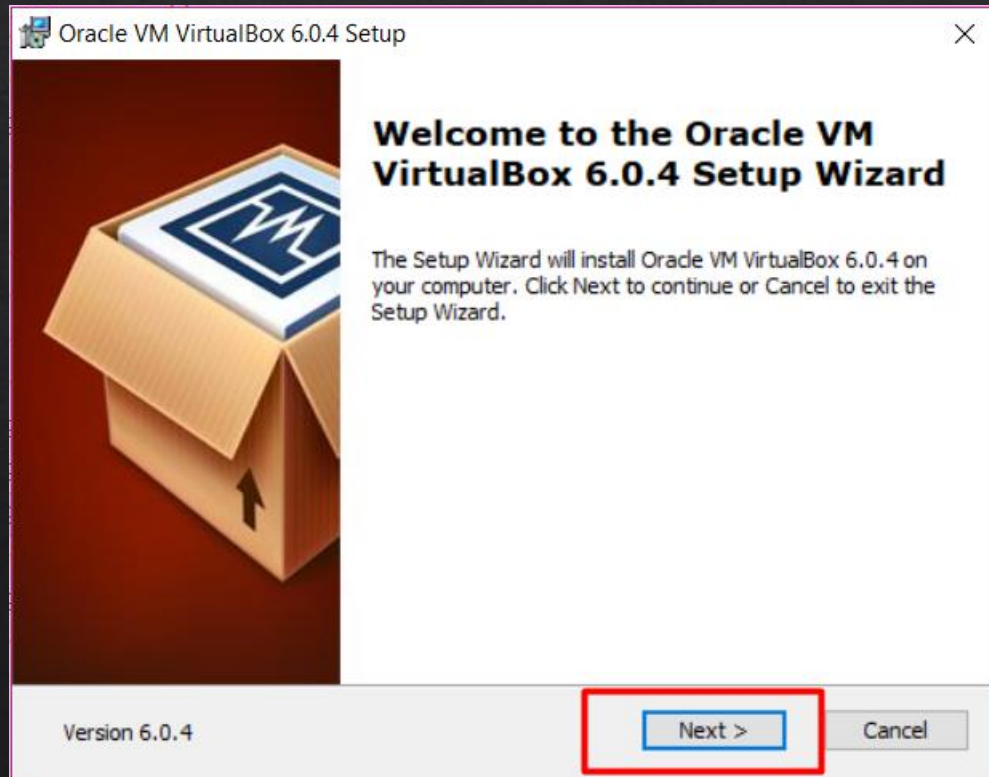
- ◊ My name is Hassan Saad.
- ◊ I'm the founder of Nine Hackers.
- ◊ I'm Information Security Consultant at Secure Networks.
- ◊ I'm Cyber Security Instructor for more than 3 years teaching a lot of certifications like:
 - ◊ OSCP
 - ◊ OSCE
 - ◊ GPEN
 - ◊ GWAPT
 - ◊ GXPN
 - ◊ GCIA
 - ◊ And others.....

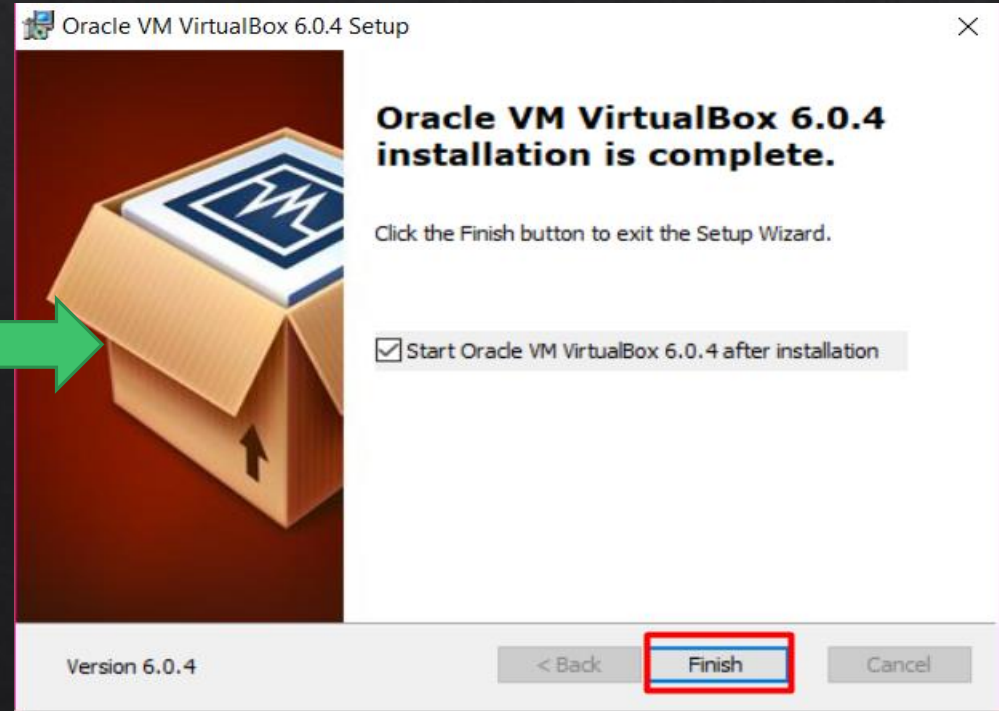
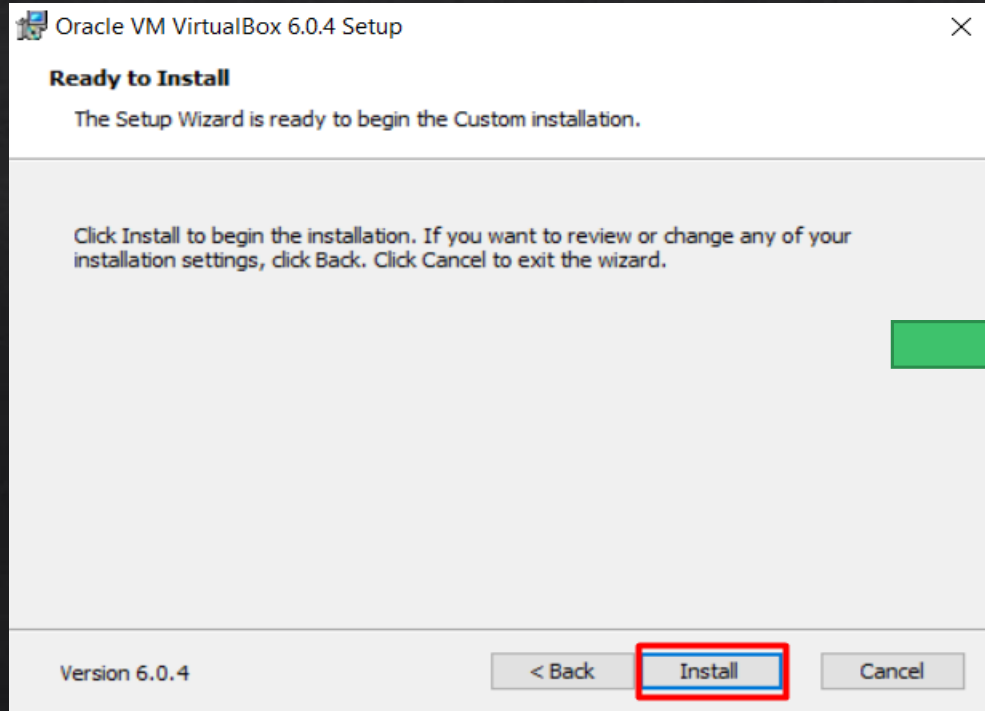
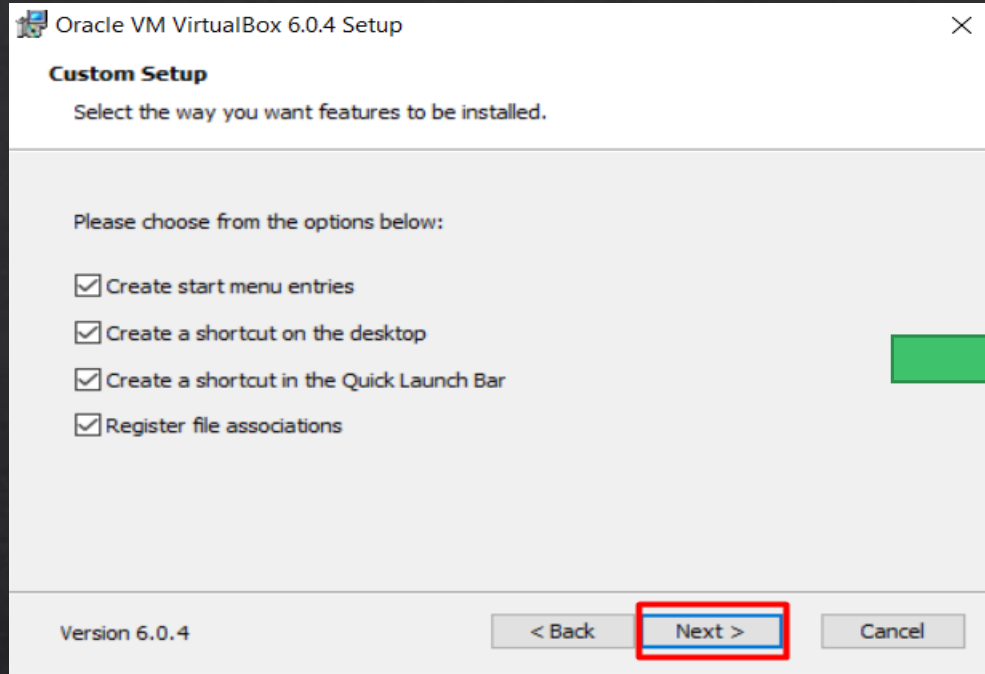
Course Materials

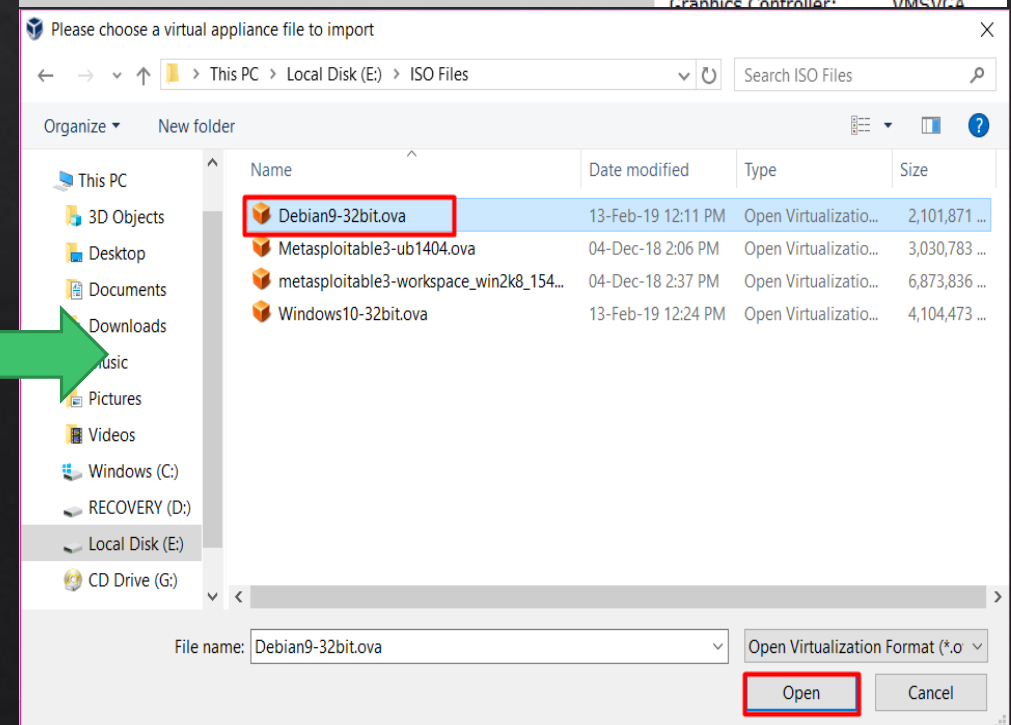
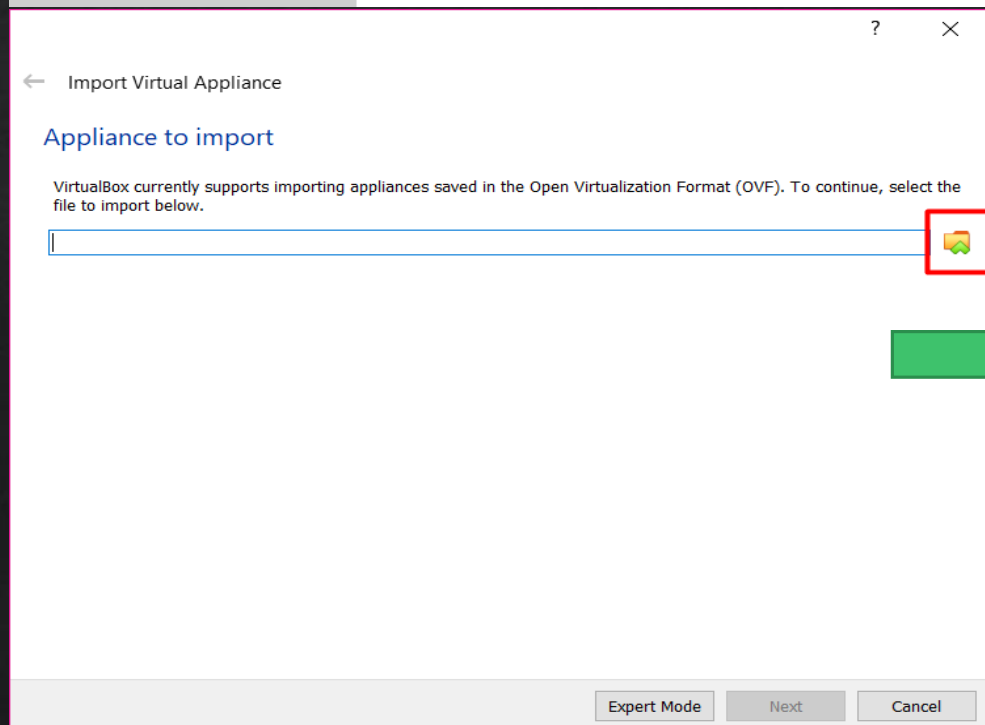
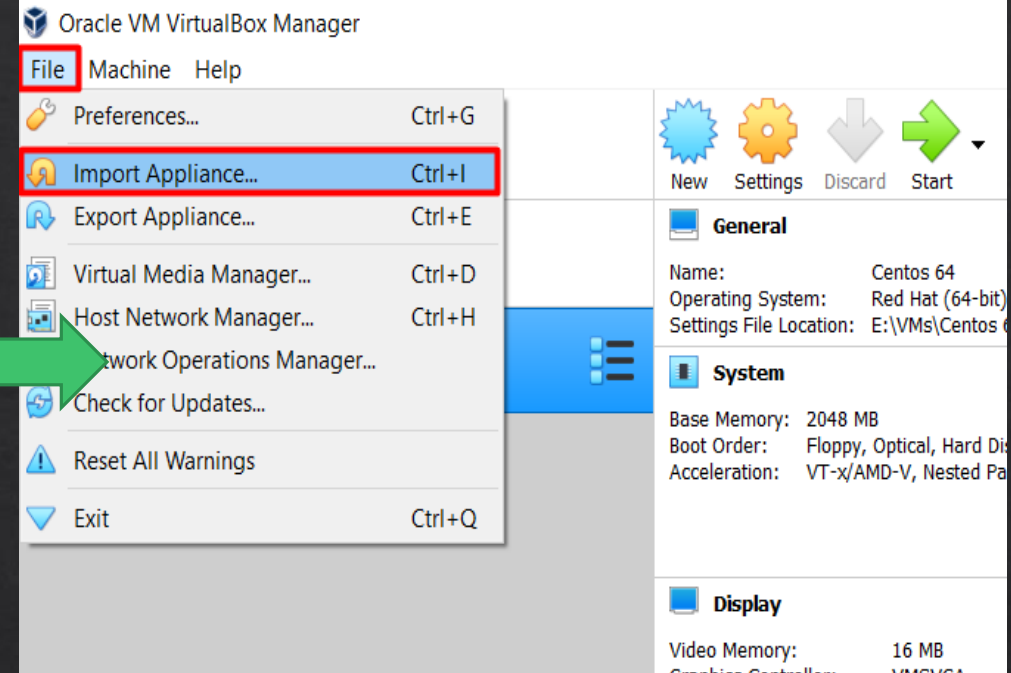
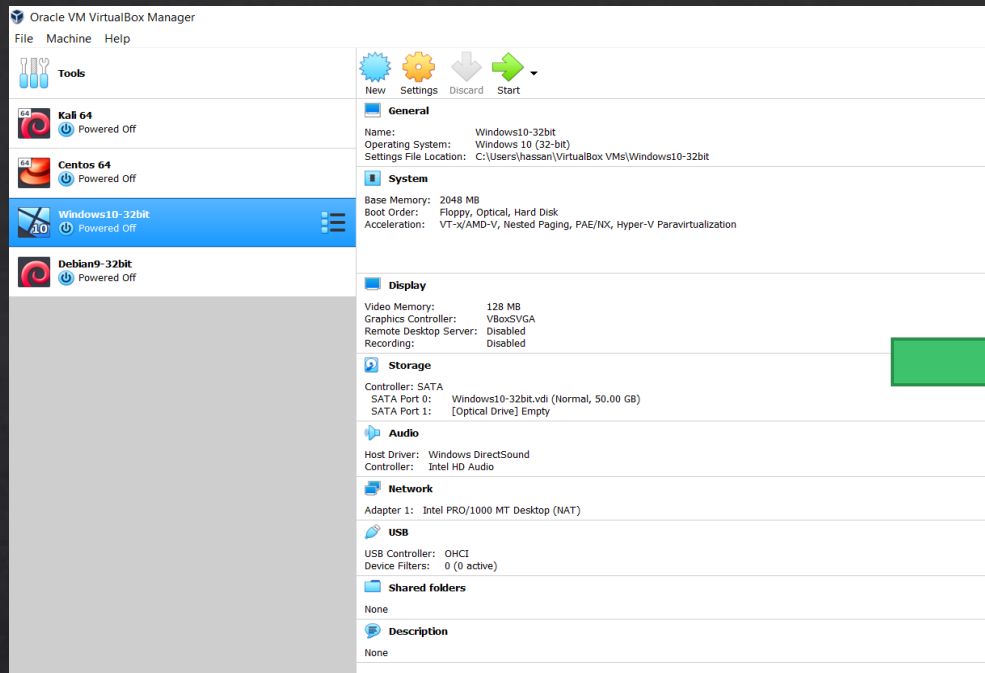
- ◇ You will find with this course folder contains this materials:
 - ◇ Books: Linux, Windows, Network, Programming.
 - ◇ Operating Systems: Debian9-32bit.ova, Windows10-32bit.ova.
 - ◇ Application: Virtualbox.exe.

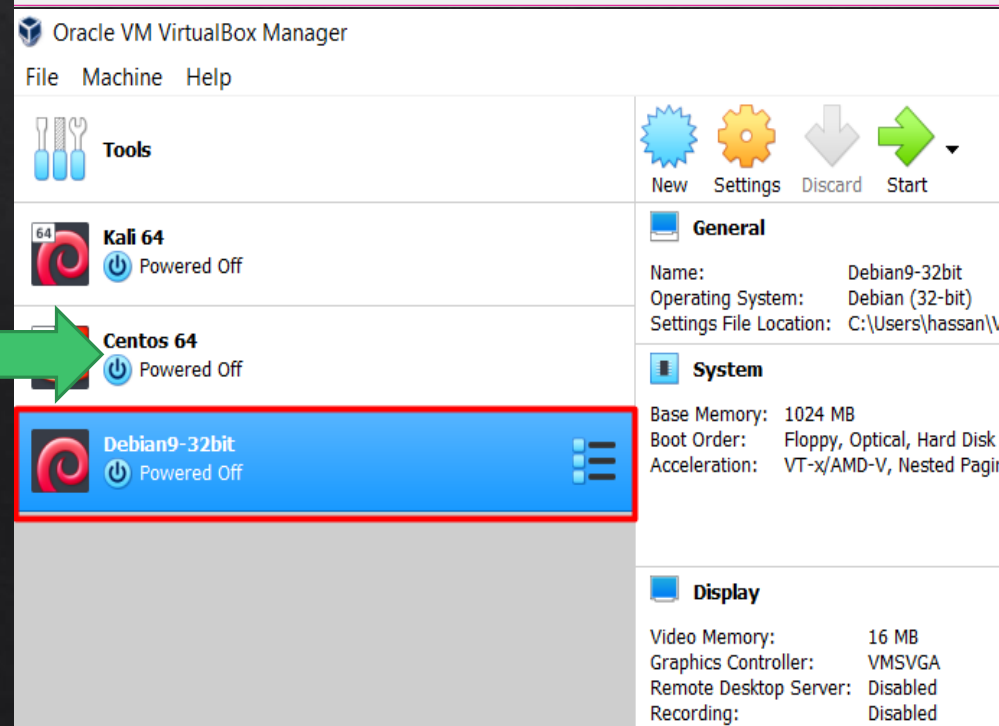
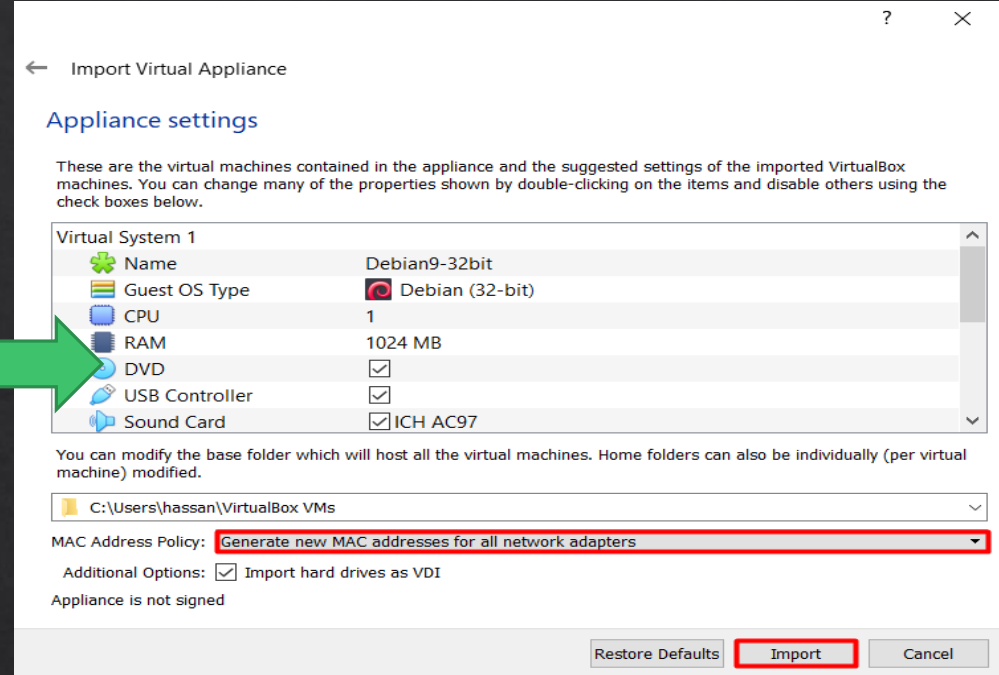
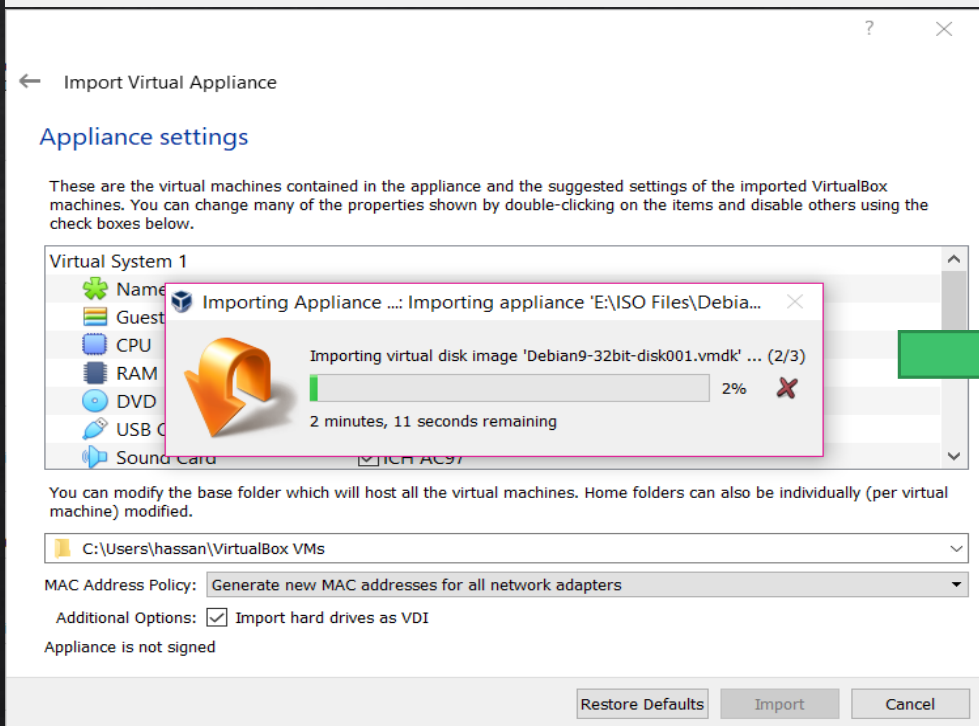
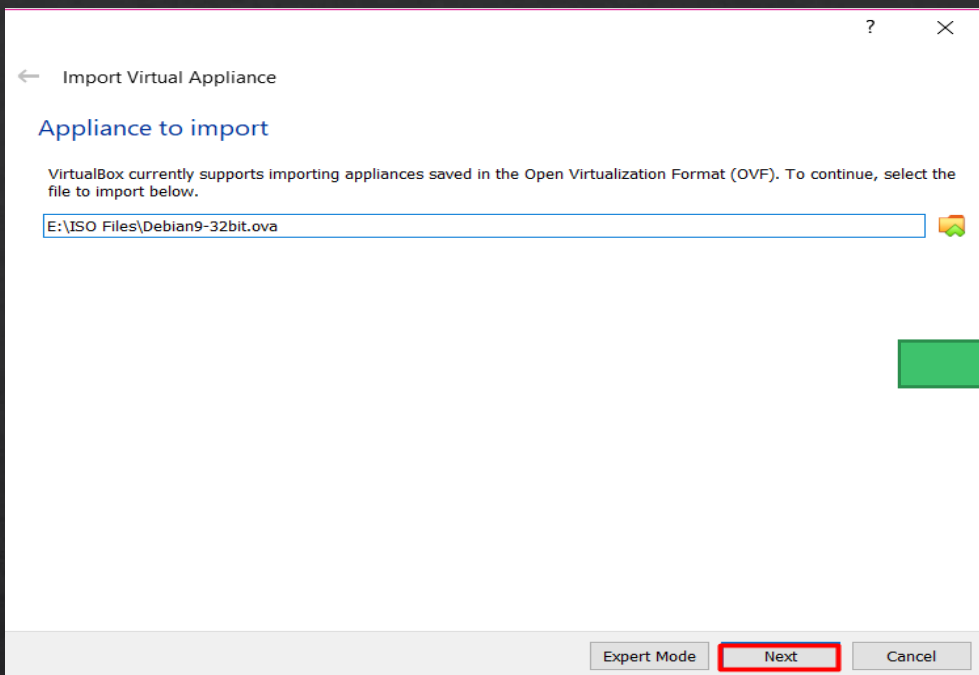
Lab Setup

- ◆ Install Virtualbox.



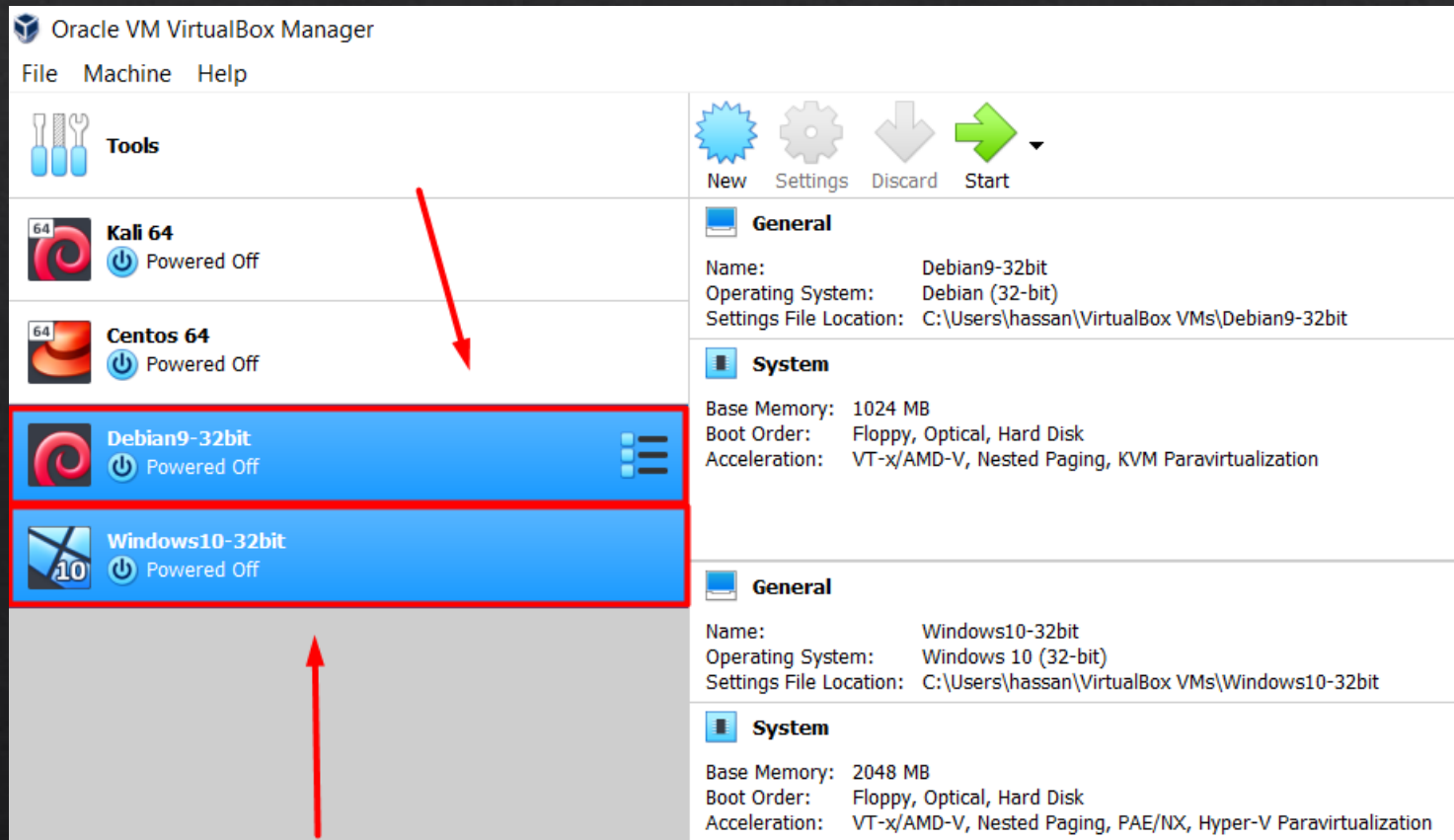




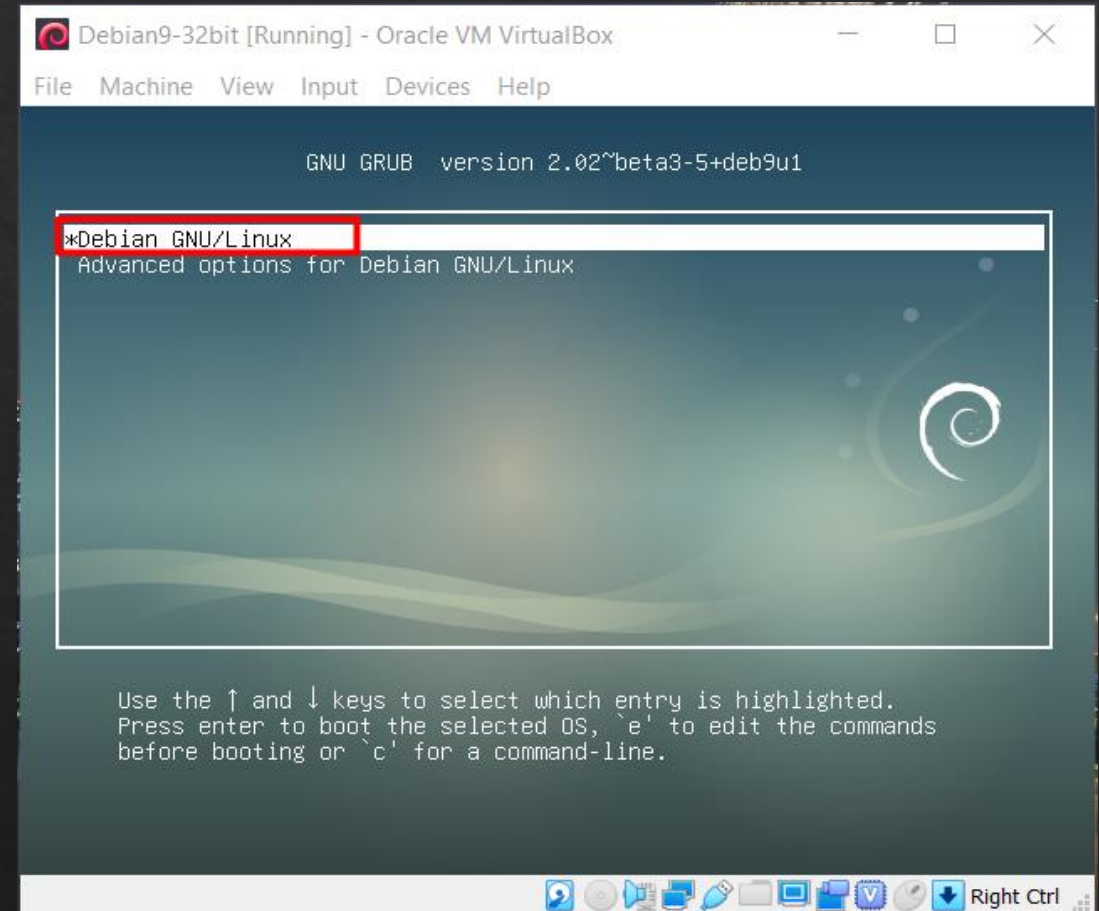
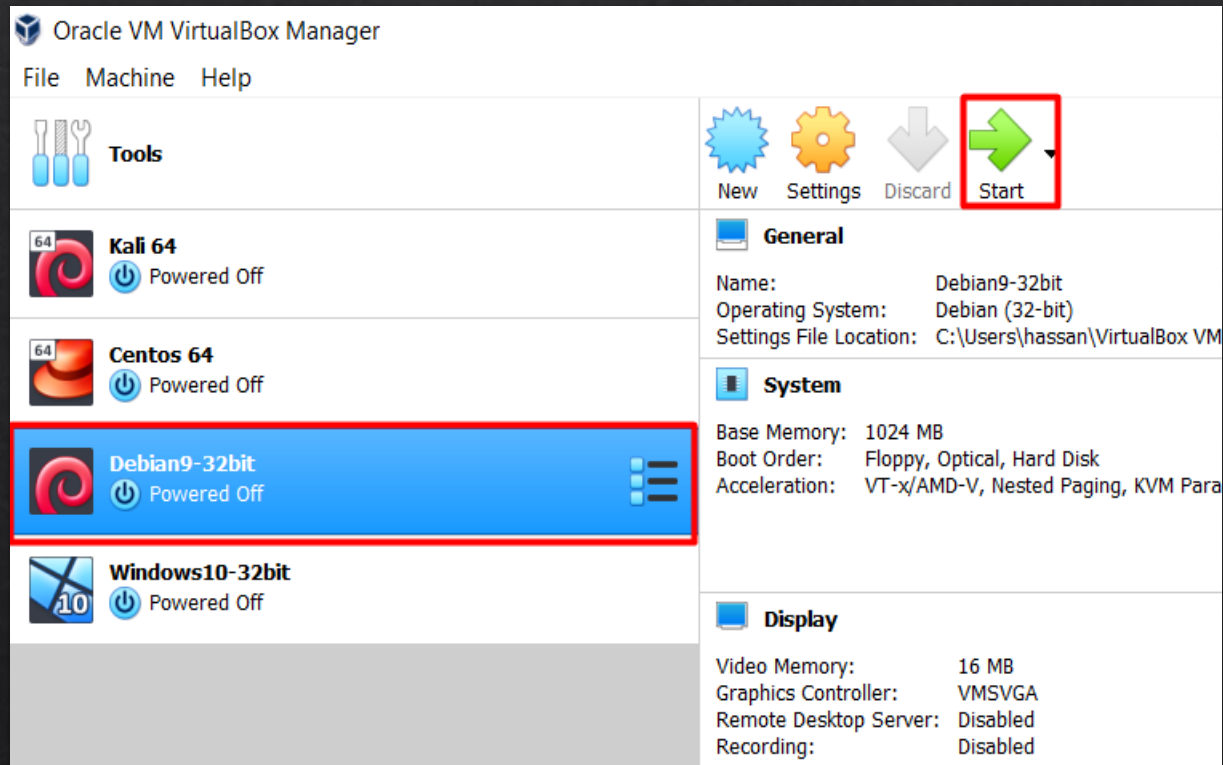


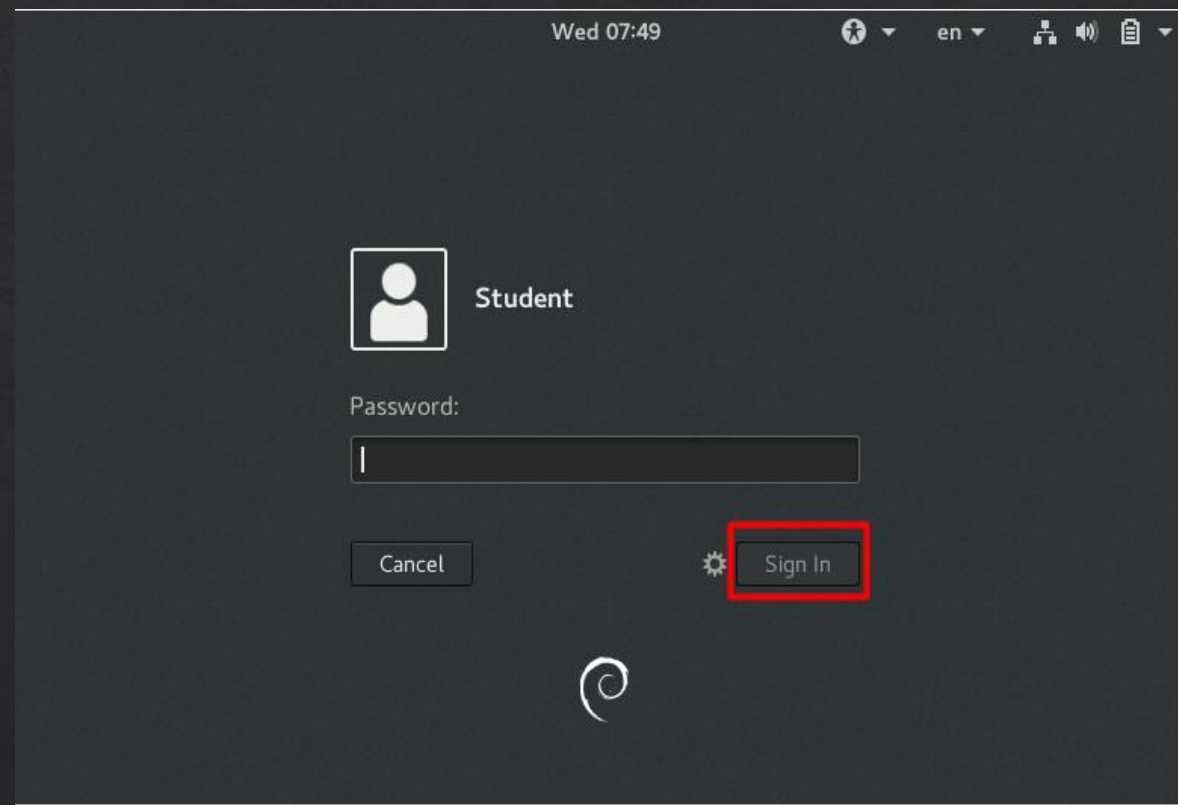
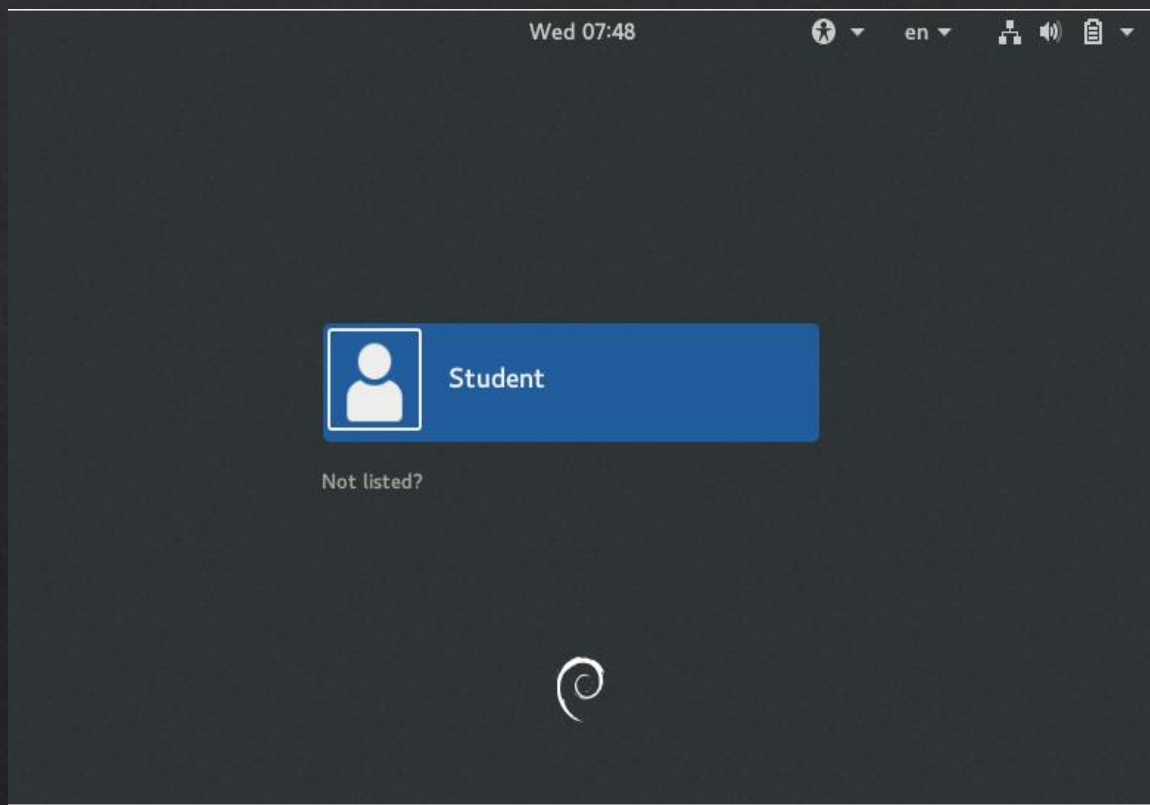
Lab Setup

- ◇ And do the same to Windows10-32bit.ova until you have both machines in Virtualbox.

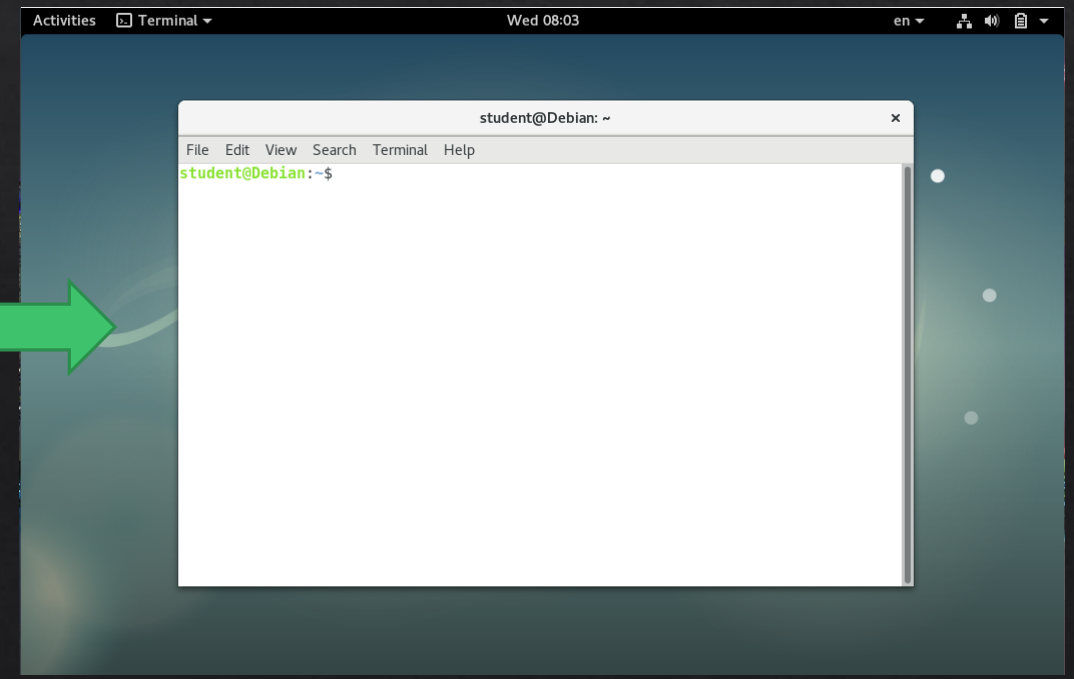
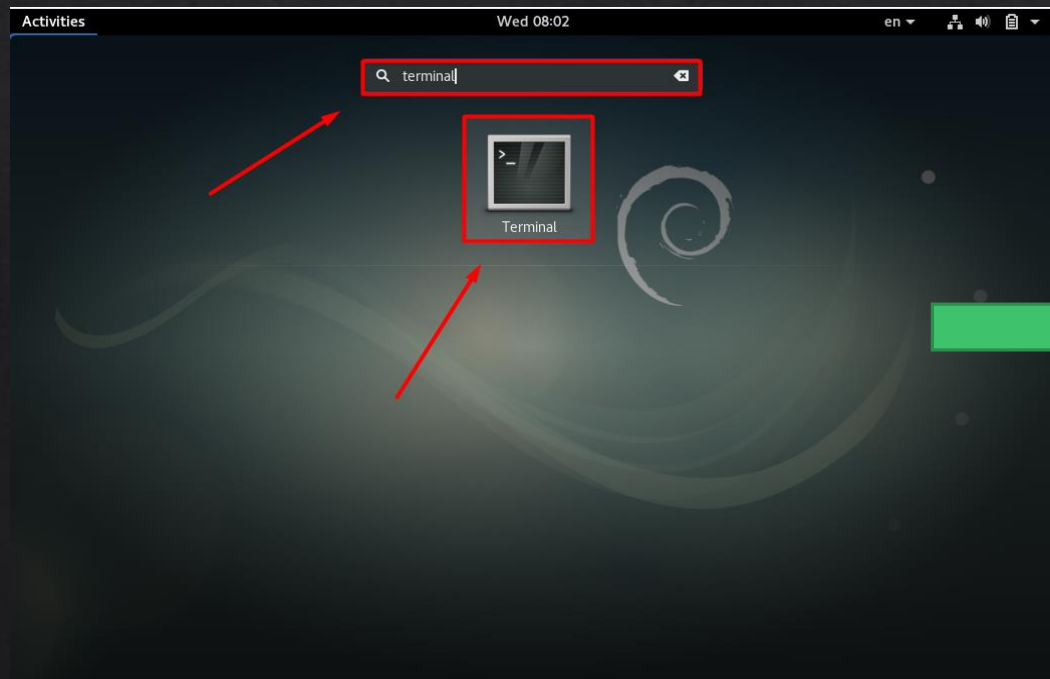
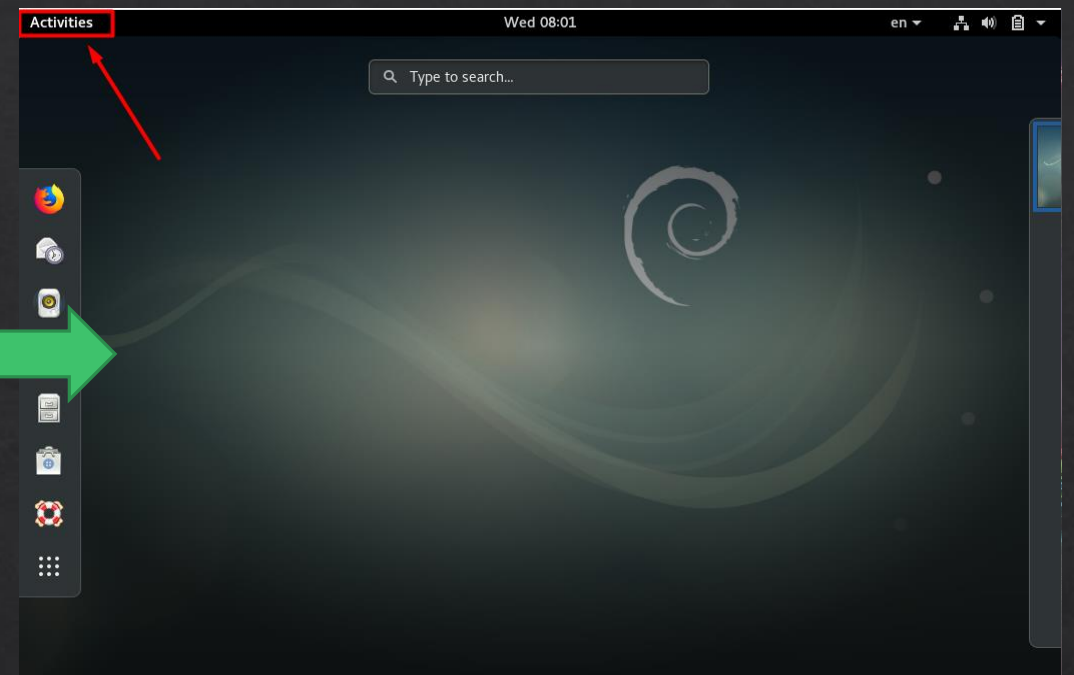
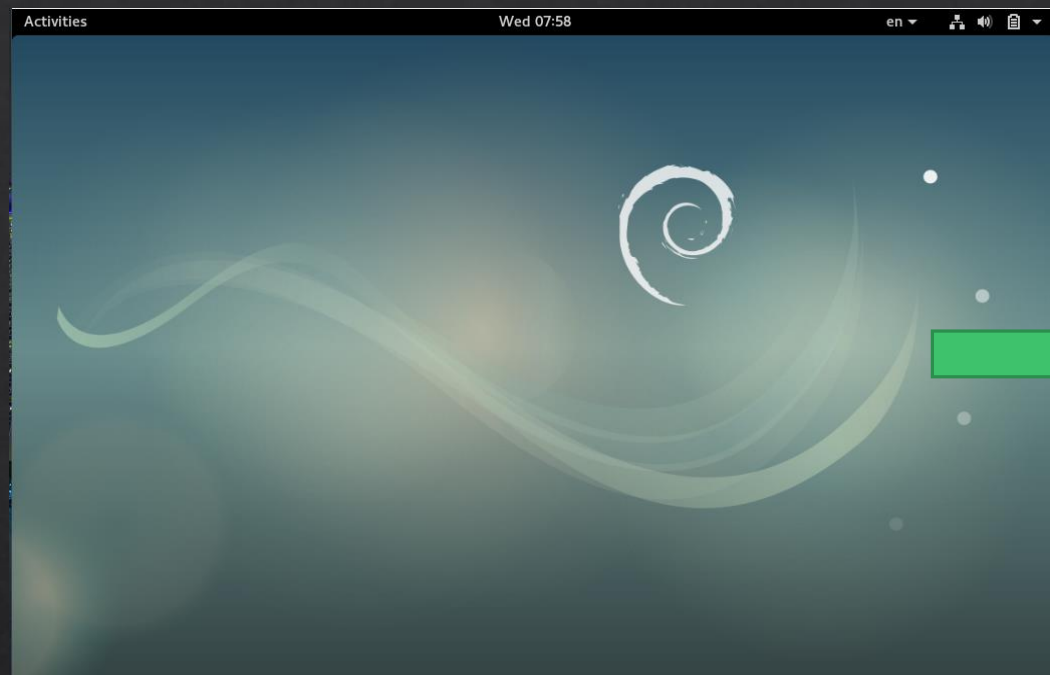


Run Debian





- ❖ Credential 1 : Username: student Password: student
- ❖ Credential 2 : Username: root Password: toor



Introduction To Linux

Core Commands

Command	Description	Example			
ls	List files and directories	ls	ls /home	ls -al	ls -al /home/student
cd	Change current directory	cd /home			
pwd	Print current directory	pwd			
cp	Copy files	cp a.txt /home			cp a.txt b.txt
mv	Move or rename file	mv a.txt /home			mv a.txt b.txt
rmdir	Remove empty directory	rmdir test			
rm	Remove file or Non empty directory	rm a.txt			rm -r test
mkdir	Create directory	mkdir test			
cat	Print file content	cat a.txt			

Core Commands

Command	Description	Example		
grep	Search for text in file	grep "word" a.txt		
head	Display the first 10 lines of a file	head a.txt	head -n 5 a.txt	
tail	Display the last 10 lines of a file	tail a.txt	tail -n 5 a.txt	tail -f a.txt
less	Display text from file in one screen	less a.txt		
ps	Display list of running processes	ps aux		
lsof	Display list of open files	lsof -i		
netstat	Display network connections	netstat -antp		
ifconfig	Display network information	ifconfig		
sort	Sort content of a file	sort a.txt		

Core Commands

Command	Description	Example	
uniq	Remove duplicate lines (sort first)	uniq a.txt	
stat	Display information about a file	stat a.txt	
ping	Test network connectivity	ping google.com	
whoami	Display current user	whoami	
passwd	Change user passwd	passwd student	
kill	Terminate process	kill 1845	
ln	Create link file	ln -s file link	ln file link
find	Search on files	find / -name a.txt	find / -name "*.txt"
nano	Text editor	nano filename (Save file : Ctrl+X)	

Special Characters

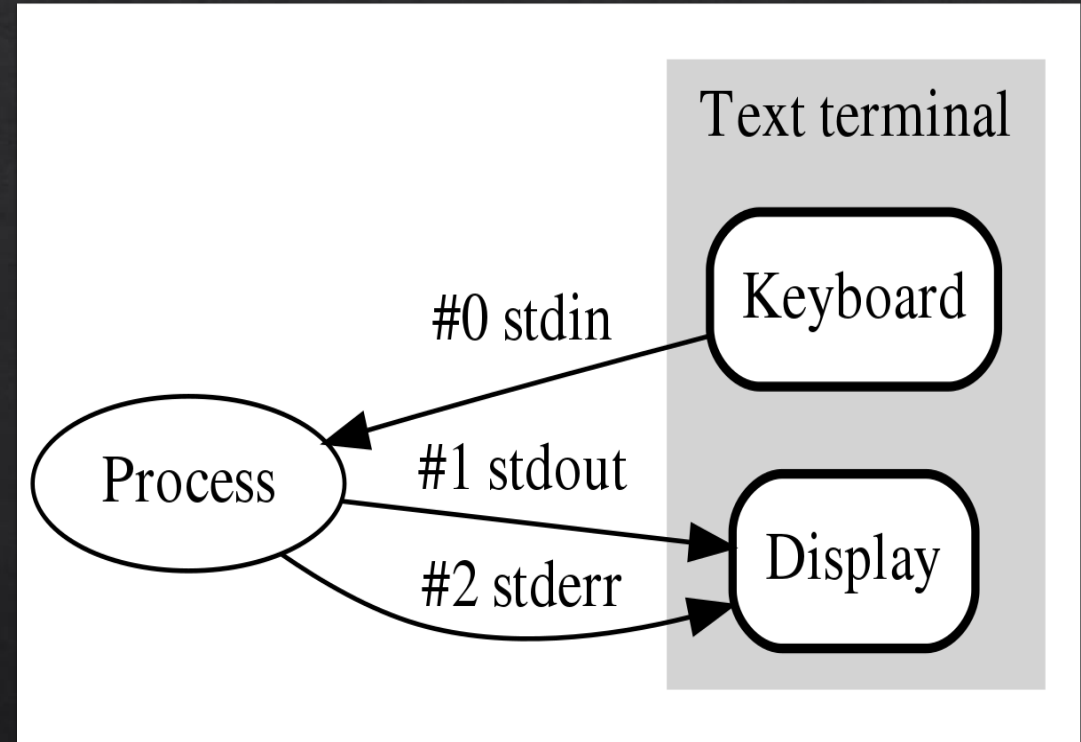
Character	Description	Example	
/	Directory separator	cd /home/student	
\	Escape character	mkdir test\ dir	
.	Current directory	ls .	cat ./a.txt
..	Parent directory	ls ..	cat ../a.txt
~	User home directory	cd ~	
&	Run in background	gedit &	
*	Represent one or more characters	ls *.txt	
?	Represent single character	ls a?.txt	
[]	Represent range of values	ls a[0-9].txt	

Special Characters

Character	Description	Example	
;	Command separator (run anyway)	pwd ; whoami	ay7aga ; whoami
&&	Command separator (run second command if the first succeed)	pwd ; whoami	ay7aga ; whoami
	Command separator (run second command if the first failed)	pwd ; whoami	ay7aga ; whoami

Redirection

- ◇ STDIN = 0.
- ◇ STDOUT = 1.
- ◇ STDERR = 2.
- ◇ `whoami > a.txt == whoami 1> a.txt`
- ◇ `ay7aga 2> err.txt`
- ◇ `wc < a.txt == wc 0< a.txt`
- ◇ `pwd >> a.txt`
- ◇ `ls > out.txt 2> err.txt`
- ◇ `ls > all.txt 2>&1`



Piping

- ◇ Pass the output of the first command to the input of the second command:
 - ◇ `ls -l | grep "Desktop"`
 - ◇ `cat /etc/passwd | grep ":0:"`
 - ◇ `ping google.com | grep "64 bytes"`
 - ◇ `cat a.txt | sort | uniq`



Environment Variables

- ◇ Run “env” command:
 - ◇ HOSTNAME=Debian.
 - ◇ USER=student.
 - ◇ PWD=/home/student.
 - ◇ PATH. (later)
 - ◇ SHELL. (later)
- ◇ echo \$HOSTNAME.
- ◇ echo \$PATH.
- ◇ echo \$SHELL.

```
PWD=/home/student
HOME=/home/student
JOURNAL_STREAM=8:16292
SSH_AGENT_PID=919
QT_ACCESSIBILITY=1
XDG_SESSION_TYPE=x11
XDG_DATA_DIRS=/usr/share/gnome:
XDG_SESSION_DESKTOP=default
GJS_DEBUG_OUTPUT=stderr
GTK_MODULES=gail:atk-bridge
WINDOWPATH=2
TERM=xterm-256color
SHELL=/bin/bash
```

PATH & SHELL Variables

- ◆ PATH is the location of your binaries.
- ◆ /usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games.
- ◆ SHELL is what type of shell you use.
- ◆ SHELL types: Bash, SH, Dash, Csh.

```
File Edit View Search Terminal Help
student@Debian:~$ echo $PATH
/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games
student@Debian:~$ PATH=$PATH:/home/student/bin
student@Debian:~$ echo $PATH
/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/home/student/bin
student@Debian:~$ PATH=/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games
student@Debian:~$ echo $PATH
/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games
```

```
File Edit View Search Terminal Help
student@Debian:~$ echo $SHELL
/bin/bash
student@Debian:~$ SHELL=/bin/sh
student@Debian:~$ echo $SHELL
/bin/sh
student@Debian:~$ $SHELL
$
$ echo $SHELL
/bin/sh
$
$ exit
student@Debian:~$ SHELL=/bin/bash
student@Debian:~$ echo $SHELL
/bin/bash
student@Debian:~$
```

User Management

- ◇ Create user:
 - ◇ useradd testuser
- ◇ Set or change user password:
 - ◇ passwd testuser
- ◇ Create group:
 - ◇ groupadd testgroup
- ◇ Add user to group:
 - ◇ gpasswd -a testuser testgroup
- ◇ Delete user:
 - ◇ userdel testuser
- ◇ Delete group:
 - ◇ groupdel testgroup

```
File Edit View Search Terminal Help
root@Debian:~# useradd testuser
root@Debian:~# useradd testuser
useradd: user 'testuser' already exists
root@Debian:~# passwd testuser
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@Debian:~# groupadd testgroup
root@Debian:~# groupadd testgroup
groupadd: group 'testgroup' already exists
root@Debian:~# gpasswd -a testuser testgroup
Adding user testuser to group testgroup
root@Debian:~# userdel testuser
root@Debian:~# userdel testuser
userdel: user 'testuser' does not exist
root@Debian:~# groupdel testgroup
root@Debian:~# groupdel testgroup
groupdel: group 'testgroup' does not exist
root@Debian:~#
```


Important Files (/etc/passwd)

- ◆ Username.
- ◆ Encrypted password.
- ◆ User ID.
- ◆ Group ID.
- ◆ Comment (Full username).
- ◆ Home directory.
- ◆ Shell type.

```
student@Debian:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
```

```
Debian-gdm:x:118:123:Gnome Display Manager:/var/lib/gdm3
student:x:1000:1000:Student,,,:/home/student:/bin/bash
vboxadd:x:999:1::/var/run/vboxadd:/bin/false
student@Debian:~$
```


Important Files (/etc/shadow)

- ◆ Username.
- ◆ Encrypted password.
 - ◆ Look at the first 3 characters.
 - ◆ \$1\$: MD5 encryption.
 - ◆ \$6\$: SHA-512 encryption.
- ◆ And some other stuff.

```
student@Debian:~$ sudo cat /etc/shadow
[sudo] password for student:
root:$6$fq5XQQCj$UHrg4kUV643WdgGp6fSLXLvy6h2Q3X9Rv1Yry8Ewb7lVUe0
2Eef7Rh6PqsiU.YrxcYfrIApmuieq/n7qwGe.:17940:0:99999:7:::
daemon:!:17940:0:99999:7:::
bin:!:17940:0:99999:7:::
```

```
Debian-gdm:!:17940:0:99999:7:::
student:$6$QubJbtXt$5EWD8Vt4yvnvu7a4o70pqUAmCzDDgtAHcRDSKo9UobULo
mTvv5DXJXDNY19Cs2afpo8QHIN069RP9It4IpD3X0:17940:0:99999:7:::
vboxadd:!:17940:0:99999:7:::
```

Important Files (/etc/group)

- ◆ Group name.
- ◆ Group password if exist.
- ◆ Group ID.
- ◆ Members of this group.

```
bluetooth:x:114:student
lpadmin:x:115:student
geoclue:x:116:
pulse:x:117:
pulse-access:x:118:
scanner:x:119:saned,student
avahi:x:120:
colord:x:121:
saned:x:122:
Debian-gdm:x:123:
student:x:1000:
vboxsf:x:999:
```

SU Command

- ◆ SU : switch to another user:
 - ◆ su
 - ◆ su testuser
- ◆ SU also can switch to another user with similar environment as the user logged in directly:
 - ◆ su -
 - ◆ su - testuser
- ◆ SU can also run command directly without full shell:
 - ◆ su username -c command

```
File Edit View Search Terminal Help
student@Debian:~$ su
Password:
root@Debian:/home/student# pwd
/home/student
root@Debian:/home/student# exit
exit
student@Debian:~$ su -
Password:
root@Debian:~# pwd
/root
root@Debian:~# exit
logout
student@Debian:~$ su testuser
Password:
$ whoami
testuser
$ exit
student@Debian:~$ su testuser -c whoami
Password:
testuser
student@Debian:~$
```

Sudo Command

- ◆ Sudo can run single command as root:
 - ◆ `sudo cat /etc/shadow`
- ◆ And you can also have full root shell:
 - ◆ `sudo -i`

```
student@Debian:~$ sudo cat /etc/shadow | tail -n 3
[sudo] password for student:
student:$6$QubJbtXT$5EWD8Vt4yvnvu7a4o70pqUAmCzDDgtAHcRDSKo9
UobULomTvv5DXJXDNY19Cs2afpo8QHiN069RP9It4IpD3X0:17940:0:999
99:7:::
vboxadd:!:17940:~::~:
testuser:$6$NwpEjDp7$NG25AGzeayA7I3E4ntxrx4nF2MH896PJXPwtw.
ODYMbftET8K1z6sMYxzonwr9mUstIPHgs22HHUFZ17P1d6j1:17940:0:99
999:7:::
student@Debian:~$ sudo -i
root@Debian:~#
root@Debian:~# whoami
root
root@Debian:~# exit
logout
student@Debian:~$
```


Linux Boot Process

- ❖ BIOS: Performs some system integrity checks then executes MBR.
- ❖ MBR: It contains information about boot loader, loads and executes the boot loader (GRUB).
- ❖ GRUB: displays a splash screen then executes kernel.
- ❖ Kernel: mount partitions then executes init script.
- ❖ Init: determine the Linux run level then executes runlevel programs.
- ❖ Runlevel: responsible for which services started at which runlevel.



Linux Run Levels

- ◈ 0 – Halt.
- ◈ 1 – Single-user text mode (rescue-mode).
- ◈ 2 – Not used (user-definable).
- ◈ 3 – Full multi-user text mode.
- ◈ 4 – Not used (user-definable).
- ◈ 5 – Full multi-user graphical mode.
- ◈ 6 – Reboot.

```
You are in rescue mode. After logging in,
system logs, "systemctl reboot" to reboot,
boot into default mode.
Give root password for maintenance
(or press Control-D to continue):
root@Debian:~#
root@Debian:~#
root@Debian:~# runlevel
N 1
root@Debian:~# systemctl get-default
rescue.target
root@Debian:~#
root@Debian:~# _
```

```
root@Debian:~# runlevel
N 3
root@Debian:~# systemctl get-default
multi-user.target
root@Debian:~#
```

```
student@Debian:~$ sudo runlevel
3 5
student@Debian:~$ sudo systemctl get-default
graphical.target
```

Run Level Commands

- ◆ Get current runlevel:
 - ◆ runlevel
- ◆ Get default runlevel:
 - ◆ systemctl get-default
- ◆ Set default runlevel:
 - ◆ systemctl set-default runlevelX.target
- ◆ List services at specific runlevel:
 - ◆ ls /etc/rcX.d
 - ◆ S means Start at boot time
 - ◆ K means Kill/Stop at boot time

```
File Edit View Search Terminal Help
student@Debian:~$ sudo runlevel
[sudo] password for student:
N 5
student@Debian:~$ sudo systemctl get-default
graphical.target
student@Debian:~$ sudo systemctl set-default runlevel3.target
Removed /etc/systemd/system/default.target.
Created symlink /etc/systemd/system/default.target → /lib/systemd/system/multi-user.target.
student@Debian:~$ sudo systemctl set-default runlevel5.target
Removed /etc/systemd/system/default.target.
Created symlink /etc/systemd/system/default.target → /lib/systemd/system/graphical.target.
student@Debian:~$ ls /etc/rc5.d/
K01apache-htcacheclean S01console-setup.sh S01gdm3 S01sneech-dispatcher
S01anacron S01cron S01minissd S01ssh
S01apache2 S01cups S01network-manager S01unattended-upgrades
S01avahi-daemon S01cups-browsed S01rsyslog
S01bluetooth S01dbus S01saned
student@Debian:~$
```

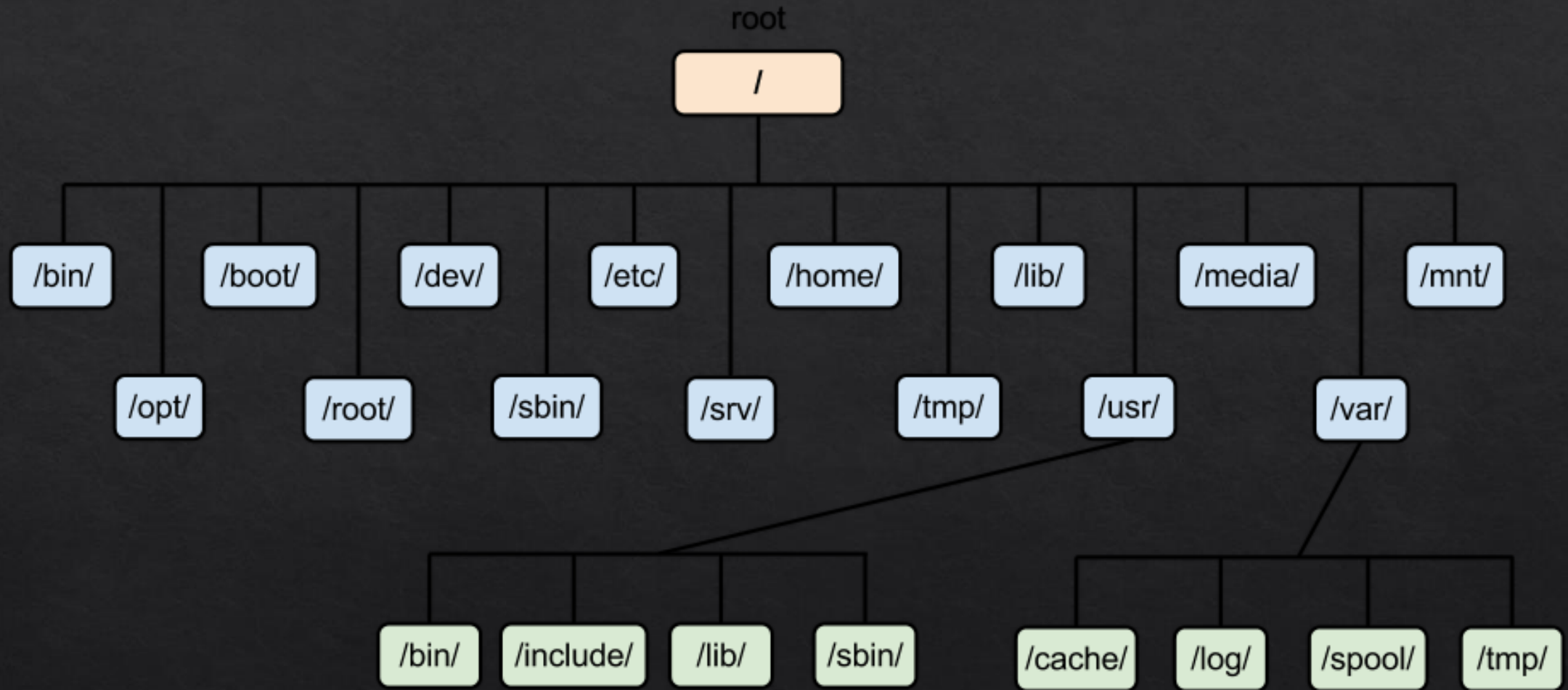
Services

- ◆ Start service:
 - ◆ systemctl start apache2
- ◆ Stop service:
 - ◆ systemctl stop apache2
- ◆ Restart service:
 - ◆ systemctl restart apache2
- ◆ Print status of service:
 - ◆ systemctl status apache2
- ◆ Enable or disable service at boot time:
 - ◆ systemctl enable/disable apache2

```
student@Debian:~$ sudo systemctl start apache2
student@Debian:~$ sudo systemctl status apache2 | grep "Active"
Active: active (running) since Wed 2019-02-13 16:11:19 EST; 1s ago
student@Debian:~$ sudo systemctl stop apache2
student@Debian:~$ sudo systemctl status apache2 | grep "Active"
Active: inactive (dead) since Wed 2019-02-13 16:11:33 EST; 2s ago
student@Debian:~$ sudo systemctl restart apache2
student@Debian:~$ sudo systemctl status apache2 | grep "Active"
Active: active (running) since Wed 2019-02-13 16:11:47 EST; 13s ago
student@Debian:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor
   Active: active (running) since Wed 2019-02-13 16:11:47 EST; 19s ago
     Process: 1835 ExecStop=/usr/sbin/apachectl stop (code=exited, status=0
     Process: 1848 ExecStart=/usr/sbin/apachectl start (code=exited, status=
   Main PID: 1852 (apache2)
```

```
student@Debian:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script
Executing: /lib/systemd/systemd-sysv-install enable apache2
student@Debian:~$
student@Debian:~$ sudo systemctl disable apache2
Synchronizing state of apache2.service with SysV service script
Executing: /lib/systemd/systemd-sysv-install disable apache2
student@Debian:~$
```


Linux File System



Linux File System

Directory name	Directory description
/	Root directory, every thing starts from there.
/root	Root home directory, contains Desktop, Downloads, Documents and so on.
/bin	Contains users binaries (ls, cp, cat).
/sbin	Contains system binaries (reboot, ifconfig, fdisk).
/etc	Contains system configuration files.
/home	Home directory for all users (/home/student, /home/testuser).
/boot	Contains boot load files and kernel files.
/lib	Contains system libraries.
/var	Contains variable data which is continuously change in size (log files: /var/log).
/usr	Contains user programs and it contains another bin (/usr/bin) & sbin (/usr/sbin) (which contains second level user and system binaries).
/mnt	Mount directory where system admin can mount any partitions here.
/tmp	Temporary files (delete at reboot).

File Permissions

- ◆ Permission groups:
 - ◆ owner(u), group(g), others(o), all(a)
- ◆ Permission types:
 - ◆ read(r=4), write(w=2), execute(x=1)
- ◆ `chmod u+rwx filename`
- ◆ `chmod g+rw filename`
- ◆ `chmod o+r filename`
- ◆ `chmod a-x filename`
- ◆ `chmod go+rw filename`

```
File Edit View Search Terminal Help
student@Debian:~$ ls -l a.txt
-rw-r--r-- 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$ chmod u+rwx a.txt
student@Debian:~$ ls -l a.txt
-rwxr--r-- 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$ chmod g+rw a.txt
student@Debian:~$ ls -l a.txt
-rwxrw-r-- 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$ chmod a+x a.txt
student@Debian:~$ ls -l a.txt
-rwxrwxr-x 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$ chmod go-wx a.txt
student@Debian:~$ ls -l a.txt
-rwxr--r-- 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$
```

File Permissions

- ◆ Permission groups:
 - ◆ owner(u), group(g), others(o), all(a)
- ◆ Permission types:
 - ◆ read(r=4), write(w=2), execute(x=1)
- ◆ `chmod 777 filename`
- ◆ `chmod 700 filename`
- ◆ `chmod 764 filename`
- ◆ `chmod 755 filename`
- ◆ `chmod 644 filename`

```
student@Debian:~$ ls -l a.txt
-rwxr--r-- 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$ chmod 777 a.txt
student@Debian:~$ ls -l a.txt
-rwxrwxrwx 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$ chmod 700 a.txt
student@Debian:~$ ls -l a.txt
-rwx----- 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$ chmod 764 a.txt
student@Debian:~$ ls -l a.txt
-rwxrw-r-- 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$ chmod 755 a.txt
student@Debian:~$ ls -l a.txt
-rwxr-xr-x 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$ chmod 644 a.txt
student@Debian:~$ ls -l a.txt
-rw-r--r-- 1 student student 75 Feb 13 09:08 a.txt
student@Debian:~$
```

Installing Software

- ◆ Update repository (always must run first):
 - ◆ `apt-get update`
- ◆ Upgrade all softwares:
 - ◆ `apt-get upgrade`
- ◆ Upgrade all the system include kernel:
 - ◆ `apt-get dist-upgrade`
- ◆ In most cases you will need to run this:
 - ◆ `apt-get update && apt-get dist-upgrade`
- ◆ Install software:
 - ◆ `apt-get install software-name`
- ◆ Remove software:
 - ◆ `apt-get remove software-name`
- ◆ Search on software:
 - ◆ `apt-cache search software-name`
- ◆ If the software don't exist in the repository then you can go to the main website of the software and download the deb package.
- ◆ Install .deb package:
 - ◆ `dpkg -i nmap.deb`
- ◆ List all packages:
 - ◆ `dpkg -l`
 - ◆ `dpkg -l | grep nmap`
- ◆ Remove software:
 - ◆ `dpkg -r nmap`